Candidate Name	Centre Number	Candidate Number

For Performance Measurement

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Advanced Level

CHEMISTRY 9189/2

PAPER 2

NOVEMBER 2017 SESSION 1 hour 15 minutes

Candidates answer on the question paper.
Additional materials:
Data Booklet
Mathematical Tables and/or Electronic Calculator

TIME 1 hour 15 minutes

INSTRUCTIONS TO CANDIDATES

Write your name, Centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.
Write your answers in the spaces provided on the question paper.

INFORMATION FOR CANDIDATES
The number of marks is given in brackets [ ] at the end of each question or part question.

FOR EXAMINER'S USE	
1	
2	
3	
4	
5	
TOTAL	

This question paper consists of 9 printed pages and 3 blank pages.

1 A standard cell was set up as shown in **Fig. 1.1**.

For Examiner's Use

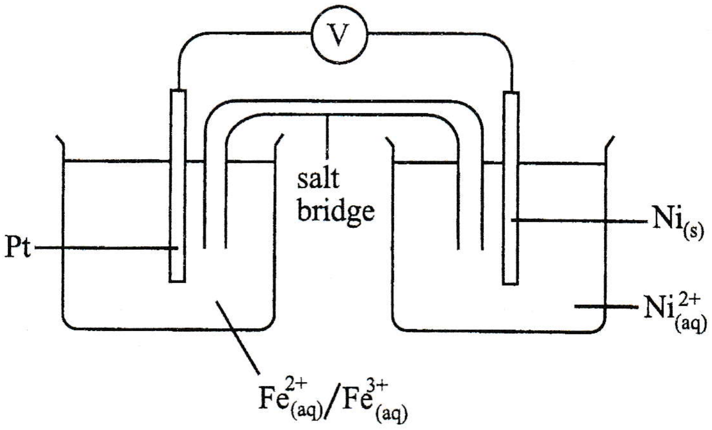

Fig. 1.1

(a) State the

(i) conditions necessary for this cell to be a standard cell,

____

____

____

____

(ii) function of the salt bridge.

____

____

____

____

[3]

(b) **(i)** Draw an arrow, on the diagram, to show the direction of electron flow in the external circuit.

For Examiner's Use

(ii) Write a half ionic equation for the reaction at the

1. cathode,

____

____

2. anode.

____

____ [3]

(c) Hence write an equation to show the overall reaction.

____

____ [1]

(d) Calculate the e.m.f. of the cell.

[1]
[Total: 8]

For Examiner's Use

2 **(a)** **(i)** Explain the difference between a strong acid and a weak acid.

(ii) Define a *Bronsted-Lowry base*.

[3]

(b) A 25.0 cm^3 sample of 0.03 $moldm^{-3}$ hydrochloric acid was diluted with 15.0 cm^3 of water.

(i) Calculate the number of moles of hydrochloric acid in the sample.

(ii) Find the concentration of hydrochloric acid in the diluted solution in mol dm^{-3}.

[3]

For Examiner's Use

(c) 25.0 cm^3 of a solution of sodium hydroxide contained 1.50×10^{-3} moles of NaOH.

Calculate the pH of this solution at 298 K.

[2]

(d) **(i)** Define the term *buffer solution*.

(ii) Give **one** example of a buffer solution.

---

[3]

[Total: 10]

For Examiner's Use

3 (a) (i) Define the term *transition element.*

(ii) Give the electronic configuration of Cr^{3+}.

[2]

(b) State the oxidation state of the transition metal in each of the following species:

(i) CrO_4^{2-}

(ii) $[CuCl_2]^-$

(iii) $[VO(H_2O)_5]^{2+}$

[3]

(c) Chromium (III) can be oxidised by hydrogen peroxide, H_2O_2.

Construct the overall equation for this reaction.

[2]

For Examiner's Use

(d) (i) Give **one** example of the use of a transition element/compound as a catalyst in a named industrial process.

(ii) Name the type of catalysis stated in **(i)**.

[3]

[Total: 10]

4 A hydrocarbon has a relative molecular mass of 106 and has the following composition by mass: C, 90.56%; H, 9.44%

(a) (i) Calculate the empirical formula of the hydrocarbon.

(ii) Deduce the molecular formula of the hydrocarbon.

For Examiner's Use

(iii) Draw structures for the **four** possible structural isomers of this hydrocarbon.

[7]

(b) In the presence of sulphuric acid, one of the four isomers, reacts with nitric acid to give only one mononitro product **Y**.

(i) Identify the isomer that reacts to produce **Y**.

(ii) Draw the structure of **Y**.

[2]
[Total: 9]

For Examiner's Use

5 Reactions 1, 2 and 3 were carried out as follows:

reaction 1 $CH_3CH_2CH_2CHO \xrightarrow{\text{one step reaction}}$ compound **A**

reaction 2 $CH_3COCH_3 \xrightarrow{\text{one step reaction}}$ compound **B**

reaction 3 compound **A** + compound **B** $\longrightarrow CH_3(CH_2)_2COOCH(CH_3)_2$

(a) Complete **Table 5.1** by stating the reagent(s), condition(s) and naming the type of the reaction.

Table 5.1

reaction	reagent(s)	conditions	type of reaction
1			
2			
3			

[9]

(b) Suggest the structural formula of compound

(i) **A**,

(ii) **B**.

[2]
[Total: 11]